



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/41

October/November 2023

2 hours

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_p = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_p = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

- 1 (a) (i) State what is indicated by the direction of the gravitational field line at a point in a gravitational field.

.....
 [1]

- (ii) Explain, with reference to gravitational field lines, why the gravitational field near the surface of the Earth is approximately constant for small changes in height.

.....

 [2]

- (b) A large isolated uniform sphere has mass M and radius R .

Point P lies on a straight line passing through the centre of the sphere, at a variable displacement x from the centre, as shown in Fig. 1.1.

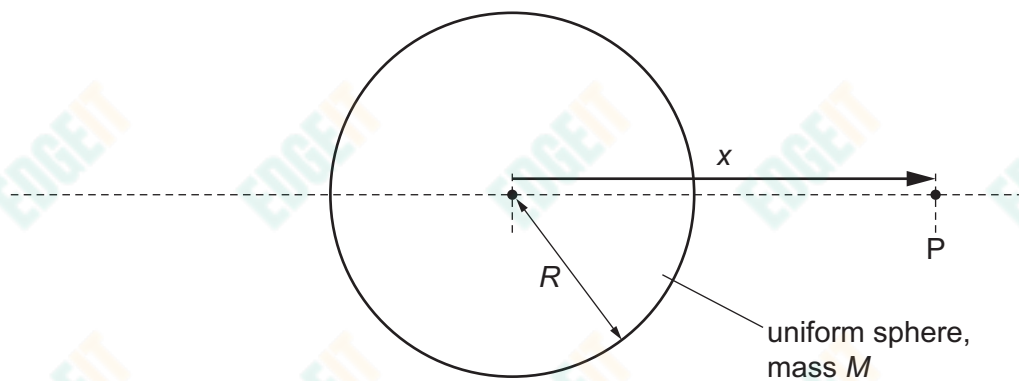


Fig. 1.1

Fig. 1.2 shows the variation with x of the gravitational field g at point P due to the sphere for the values of x for which P is inside the sphere.

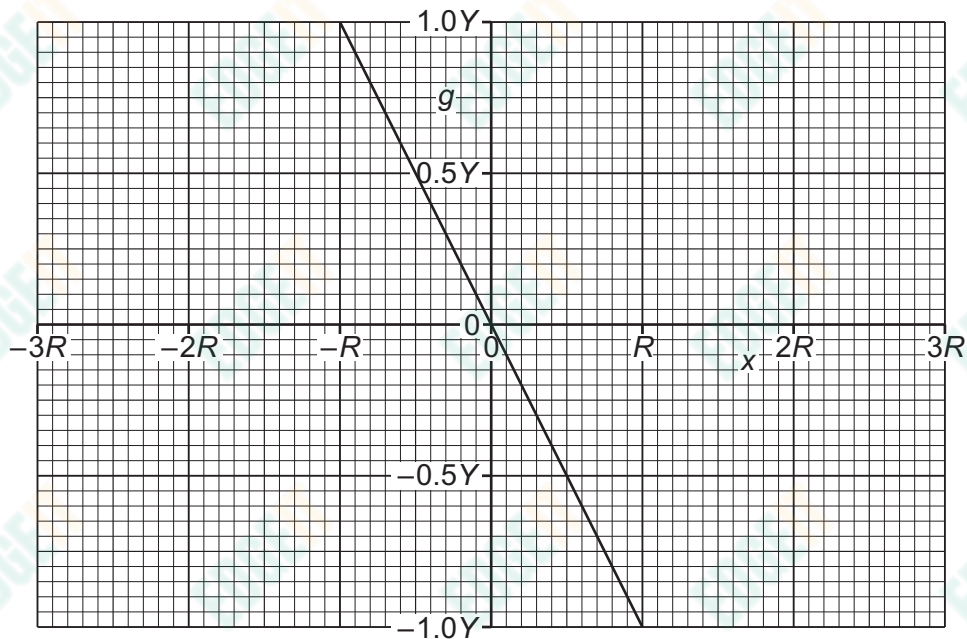


Fig. 1.2

The magnitude of the gravitational field at the surface of the sphere is Y .

- (i) Determine an expression for Y in terms of M and R . Identify any other symbols that you use.

[2]

- (ii) Explain why, at the surface of the sphere, g always has the opposite sign to x .

.....

.....

..... [2]

- (iii) Complete Fig. 1.2 to show the variation of g with x for values of x , up to $\pm 3R$, for which point P is outside the sphere. [3]

[Total: 10]

- 2 (a) Define specific heat capacity.

.....

 [2]

- (b) An ideal gas of mass 0.35 kg is heated at a constant pressure of $2.0 \times 10^5\text{ Pa}$ so that its internal energy increases by 7600 J . During this process, the volume of the gas increases from 0.038 m^3 to 0.063 m^3 and the temperature increases by 56°C .

- (i) Show that the magnitude of the work done on the gas is 5000 J .

[1]

- (ii) Explain whether the work done on the gas is positive or negative.

.....

 [2]

- (iii) Determine the magnitude of the thermal energy q transferred to the gas.

$$q = \dots\dots\dots \text{ J} [2]$$

- (iv) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

$$\text{specific heat capacity} = \dots\dots\dots \text{ unit} \dots\dots\dots [2]$$

- (c) The gas in (b) is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in (b).

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in (b)(iv).

.....

.....

.....

.....

..... [3]

[Total: 12]

- 3 (a) The product pV for an ideal gas is given by

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where p is the pressure of the gas and V is the volume of the gas.

- (i) State the meaning of the symbols N , m and $\langle c^2 \rangle$ in this equation.

N :

m :

$\langle c^2 \rangle$:

[3]

- (ii) Use the equation of state for an ideal gas to show that the average translational kinetic energy E_K of a molecule of the gas at thermodynamic temperature T is given by

$$E_K = \frac{3}{2} kT.$$

[2]

- (b) The surface of a star consists mainly of a gas that may be assumed to be ideal. The molecules of the gas have a root-mean-square (r.m.s.) speed of 9300 m s^{-1} .

The mass of a molecule of the gas is $3.34 \times 10^{-27} \text{ kg}$.

Determine, to three significant figures, the temperature of the surface of the star.

temperature = K [2]

- (c) The radiant flux intensity of the radiation from the star in (b) is $2.52 \times 10^{-8} \text{ W m}^{-2}$ when observed at a distance of $4.16 \times 10^{16} \text{ m}$ from the star.

(i) Calculate the luminosity of the star. Give a unit with your answer.

luminosity = unit [2]

(ii) Determine the radius of the star.

radius = m [2]

- (d) The gas at the surface of a star has a very high pressure.

Use the basic assumptions of the kinetic theory to suggest why, in practice, a gas at the surface of a star is unlikely to behave as an ideal gas.

.....

.....

.....

..... [2]

[Total: 13]

- 4 A heavy metal sphere of mass 0.81 kg is suspended from a string. The sphere is undergoing small oscillations from side to side, as shown in Fig. 4.1.

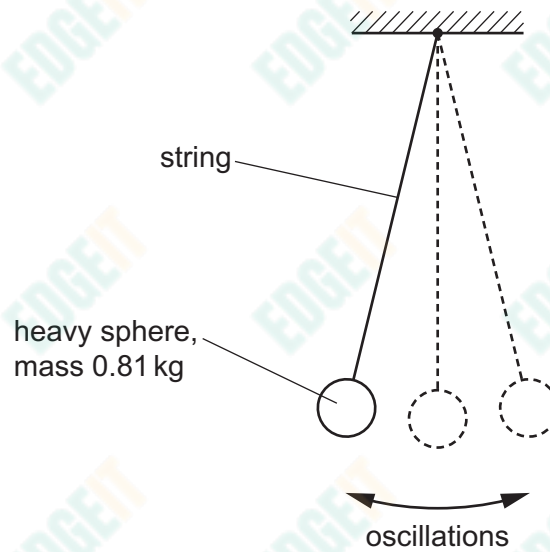


Fig. 4.1

The oscillations of the sphere may be considered to be simple harmonic with amplitude 0.036 m and period 3.0 s.

- (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) Calculate:

- (i) the angular frequency of the oscillations

angular frequency = rad s^{-1} [2]

- (ii) the total energy of the oscillations.

total energy = J [2]

- (c) The suspended sphere is now lowered into water. The sphere is given a sideways displacement of $+0.036\text{ m}$ from its equilibrium position and is then released at time $t = 0$. The water causes the motion of the sphere to be critically damped.

On Fig. 4.2, sketch the variation of the displacement x of the sphere from its equilibrium position with t from $t = 0$ to $t = 6.0\text{ s}$.

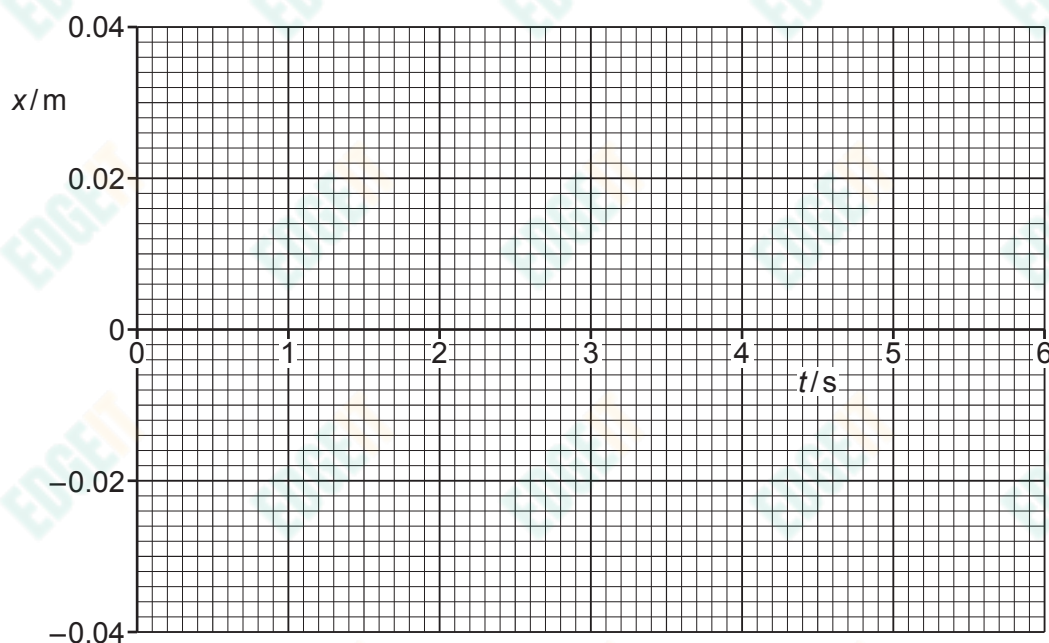


Fig. 4.2

[3]

[Total: 9]

- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance x from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with x of the electric potential V due to the spheres at point P.

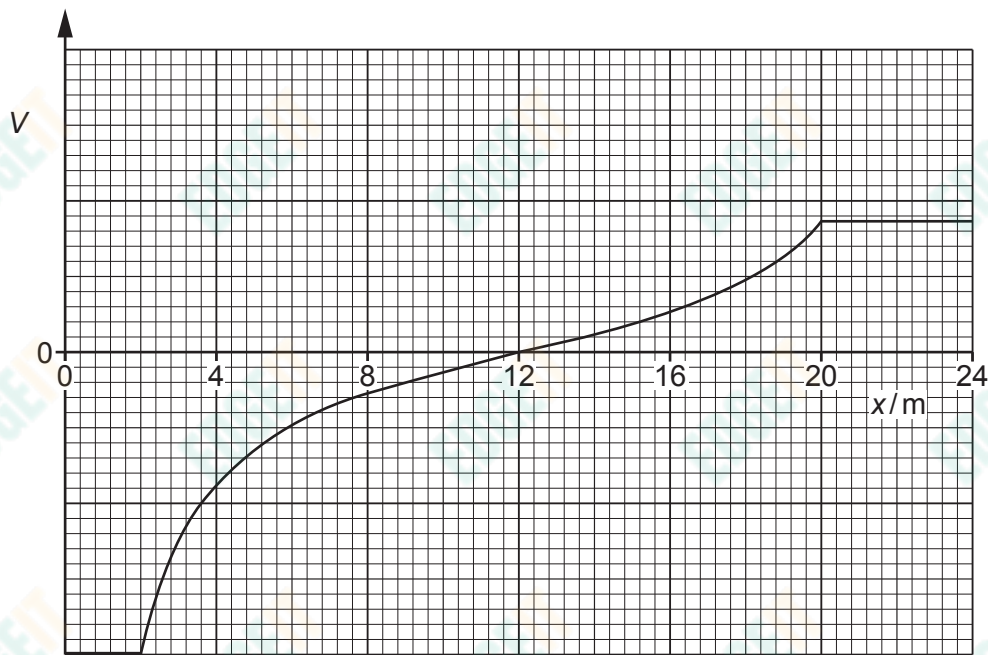


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) A positively charged particle is placed at point P in (b), such that $x = 12\text{ m}$. The particle is released.

Describe and explain the subsequent motion of the particle.

.....

.....

.....

.....

..... [3]

[Total: 8]

- 6 (a) Define magnetic flux density.

.....

.....

..... [2]

- (b) Electrons are moving in a vacuum with speed $1.7 \times 10^7 \text{ ms}^{-1}$. The electrons enter a uniform magnetic field of flux density 4.8 mT . Fig. 6.1 shows the path of the electrons.

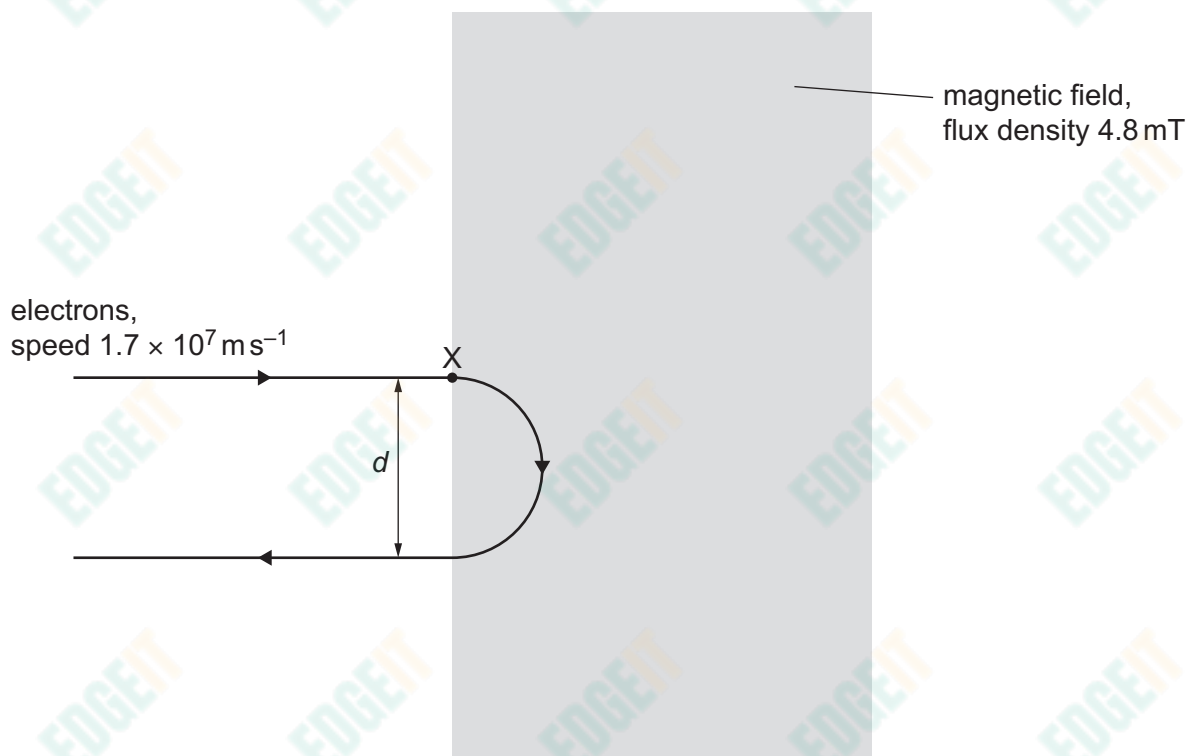


Fig. 6.1

The path of the electrons remains in the plane of the page.

- (i) State the direction of the magnetic field.

.....

..... [1]

- (ii) Show that the magnitude of the force exerted on each electron by the magnetic field is $1.3 \times 10^{-14} \text{ N}$.

[2]

- (iii) On Fig. 6.1, draw an arrow to indicate the direction of the centripetal acceleration of the electron where it enters the magnetic field at point X. [1]

- (iv) Use the information in **(b)(ii)** to calculate the distance d between the path of the electrons entering the magnetic field and the path of the electrons leaving it.

$d = \dots\dots\dots \text{ m}$ [3]

- (c) The electrons in **(b)** are replaced with positrons that are moving with speed $3.4 \times 10^7 \text{ ms}^{-1}$ along the same initial path as the electrons.
The positrons enter the magnetic field at point X on Fig. 6.1.

On Fig. 6.1, draw a line to show the path of the positrons through the magnetic field. [3]

[Total: 12]

- 7 A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

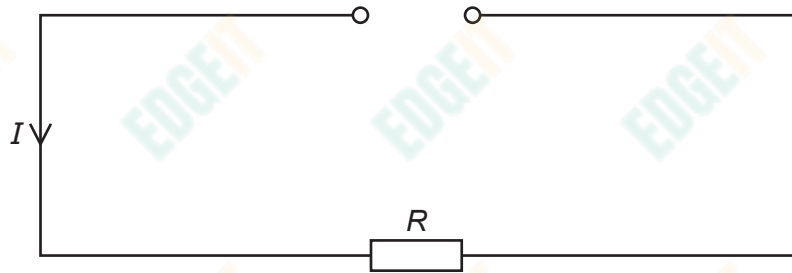


Fig. 7.1

Fig. 7.2 shows the variation with time t of I .

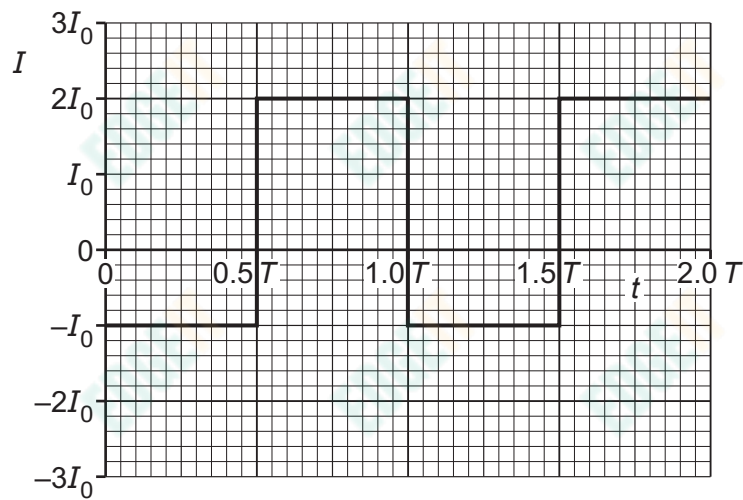


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T .

- (a) Determine expressions, in terms of I_0 and R , for the power P dissipated in the resistor for the times when:

- (i) the current is in the negative direction

$$P = \dots\dots\dots [1]$$

- (ii) the current is in the positive direction.

$$P = \dots\dots\dots [1]$$

- (b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

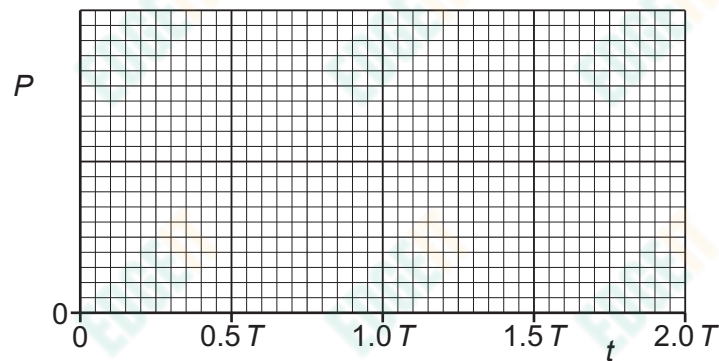


Fig. 7.3

[3]

- (c) Use your answer in (b) to determine an expression, in terms of I_0 and R , for:

- (i) the mean power $\langle P \rangle$ in the resistor

$$\langle P \rangle = \dots\dots\dots [1]$$

- (ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$$I_{\text{r.m.s.}} = \dots\dots\dots [2]$$

[Total: 8]

- 8 (a) (i) Show that the momentum p of a photon of electromagnetic radiation with wavelength λ is given by

$$p = \frac{h}{\lambda}$$

where h is the Planck constant.

[2]

- (ii) Use the expression in (a)(i) to show that a photon in free space that has a momentum of $9.5 \times 10^{-28} \text{ N s}$ is a photon of red light.

[1]

- (b) A beam of red light of intensity 160 W m^{-2} is incident normally on a plane mirror, as shown in Fig. 8.1. The momentum of each photon in the beam is $9.5 \times 10^{-28} \text{ N s}$.

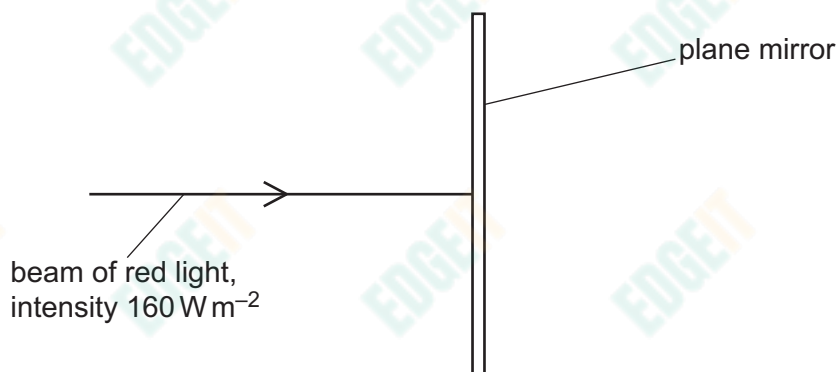


Fig. 8.1

All of the light is reflected by the mirror in the opposite direction to its original path. The cross-sectional area of the beam is $2.5 \times 10^{-6} \text{ m}^2$.

- (i) Show that the number of photons incident on the mirror per unit time is $1.4 \times 10^{15} \text{ s}^{-1}$.

[2]

- (ii) Use the information in **(b)(i)** to determine the pressure exerted by the light beam on the mirror.

pressure = Pa [3]

- (c) The beam of red light in **(b)** is now replaced with a beam of blue light of the same intensity.

Suggest and explain whether the pressure exerted on the mirror by the beam of blue light is less than, the same as, or greater than the pressure exerted by the beam of red light.

.....

.....

.....

..... [2]

[Total: 10]

- 9 (a) State what is meant by nuclear fusion.

.....

.....

..... [2]

- (b) On Fig. 9.1, sketch the variation of binding energy per nucleon with nucleon number A for values of A between 1 and 250.



Fig. 9.1

- (c) On your line in Fig. 9.1, label:

- (i) a point X that could represent a nucleus that undergoes alpha-decay [1]
- (ii) a point Y that could represent a nucleus that undergoes nuclear fusion. [1]

- (d) A nucleus Z undergoes nuclear fission to form strontium-93 (${}^{93}_{38}\text{Sr}$) and xenon-139 (${}^{139}_{54}\text{Xe}$) according to

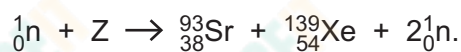


Table 9.1 shows the binding energies of the strontium-93 and xenon-139 nuclei.

Table 9.1

nucleus	binding energy/J
${}^{93}_{38}\text{Sr}$	1.25×10^{-10}
${}^{139}_{54}\text{Xe}$	1.81×10^{-10}

The fission of 1.00 mol of Z releases 1.77×10^{13} J of energy.

Determine the binding energy per nucleon, in MeV, of Z.

binding energy per nucleon = MeV [4]

[Total: 10]

- 10 Ultrasound and X-rays are both types of wave that are used in medical diagnosis to form images of internal body structures.

(a) Complete Table 10.1 to state, for each type of wave:

- the method of production of the wave
- whether the wave that is detected and used to form the image is the wave that has been absorbed, reflected or transmitted by the internal body structure.

Table 10.1

	ultrasound	X-rays
method of production		
detected wave (absorbed, reflected or transmitted)		

[4]

- (b) (i) For one type of wave passing through tissue, the wave has 72% of its initial intensity after it has passed through 6.2 cm of the tissue.

Calculate the linear attenuation coefficient μ of the tissue for this wave.

$$\mu = \dots\dots\dots \text{cm}^{-1} \quad [2]$$

- (ii) Another wave of the same type as in (b)(i) passes through 9.3 cm of the same tissue.

Calculate the percentage of the initial intensity of the wave that is attenuated by the tissue.

$$\text{percentage attenuated} = \dots\dots\dots \% \quad [2]$$

[Total: 8]

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